

# AP Statistics

## AP Վիճակագրություն

|              |                                                                                       |
|--------------|---------------------------------------------------------------------------------------|
| COURSE CODE  | MTH-APS                                                                               |
| PROGRAM      | Advanced Placement and Examination Preparation · Առաջադեմ և քննական պատրաստություն    |
| LEVEL        | Typically Grades 10–12 · College Board aligned                                        |
| DURATION     | 126–144 instructional hours (approx. 100 lessons) plus examination preparation        |
| PREREQUISITE | Algebra II (MTH-A2) or equivalent. Calculus is not required.                          |
| LEADS TO     | University statistics, research methods, or quantitative coursework in any discipline |

### COURSE DESCRIPTION

Դասընթացի նկարագրություն

AP Statistics is equivalent to a one-semester introductory university statistics course and is organized around nine College Board units. It covers exploring one- and two-variable data, collecting data through sampling and experimental design, probability and random variables, sampling distributions, and inference for proportions, means, categorical data, and regression slopes. The course is built around four skill categories: selecting statistical methods, describing data and distributions, using probability and simulation, and — most heavily assessed — arguing from evidence in writing.

### TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

The AP Statistics examination is a writing examination that happens to be about statistics. Roughly half the score comes from free-response questions scored on whether the student identified the correct procedure, verified its conditions, executed it, and stated a conclusion in context with linkage to the evidence. Վարժարան therefore trains a four-step written template from Unit 1 and requires it on every inference problem for the rest of the course. Computation is deliberately delegated to technology: the assessed skill is choosing correctly, checking honestly, and communicating precisely. A student who can run a test but cannot write the conclusion has learned the least valuable half of the subject.

### PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

| READINESS INDICATOR                                                                          | WHAT IT TELLS THE TEACHER                                                                        |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Computes with proportions, percents, and decimals fluently</b>                            | Sufficient arithmetic for the whole course; heavy algebra is not required.                       |
| <b>Reads and interprets graphs, and understands slope in context</b>                         | Prerequisite for Unit 2 regression work.                                                         |
| <b>Writes a clear paragraph explaining a quantitative claim</b>                              | The most predictive readiness indicator for this course, and the one schools least often assess. |
| <b>Uses a graphing calculator's statistical menus, or is willing to learn them in Unit 1</b> | Distributions, regression, and inference procedures are all calculator-executed.                 |

## COURSE LEARNING OUTCOMES

Ուսումնական արդյունքներ

On completion of this course the student will be able to:

- 1 Describe and compare distributions of one-variable data using shape, centre, spread, and unusual features.
- 2 Use the normal distribution and z-scores to compute proportions and percentiles.
- 3 Analyze two-variable data with scatter plots, correlation, and least-squares regression, including residual analysis.
- 4 Identify and describe sampling methods, sources of bias, and the scope of inference.
- 5 Design experiments applying control, randomization, and replication, including blocking.
- 6 Apply probability rules including conditional probability and independence.
- 7 Work with random variables, expected value, and the binomial and geometric distributions.
- 8 Describe sampling distributions and apply the Central Limit Theorem.
- 9 Construct and interpret confidence intervals for proportions and means, single and two-sample.
- 10 Carry out significance tests for proportions and means, single and two-sample, including paired data.
- 11 Perform chi-square goodness of fit, homogeneity, and independence tests.
- 12 Perform inference for the slope of a least-squares regression line.
- 13 Verify the conditions for every inference procedure before applying it.
- 14 State conclusions in context with explicit linkage to the evidence.

## COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

### UNIT 1 · 16-17 LESSONS · APPROX. 15-23% OF THE EXAMINATION

## Exploring One-Variable Data

Միաչափ տվյալների ուսումնասիրություն

### TOPICS COVERED

- Introducing statistics: what can we learn from data?
- The language of variation: variables
- Representing a categorical variable with tables and graphs
- Representing a quantitative variable with graphs
- Describing the distribution of a quantitative variable
- Summary statistics for a quantitative variable
- Graphical representations of summary statistics
- Comparing distributions of a quantitative variable
- The normal distribution and its parameters
- The empirical rule and z-scores
- Normal distribution calculations in both directions

### LEARNING OBJECTIVES

- 1 Describe a distribution completely in context: shape, centre, spread, and unusual features.
- 2 Compare two distributions using explicitly comparative language.
- 3 Compute and interpret z-scores and use the normal distribution in both directions.
- 4 Select an appropriate measure of centre and spread and justify the choice from the shape.

### MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces complete contextual descriptions in 9 of 10 tasks.
- Uses comparative language correctly when comparing distributions in 9 of 10 trials.
- Performs normal calculations at 92% accuracy in both directions.

### COMMON MISCONCEPTIONS AND TEACHER RESPONSE

| MISCONCEPTION                                                                  | RECOMMENDED RESPONSE                                                                                                              |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Distributions are described without context or with an element missing.</b> | All four elements, always, with variable names and units. This is a scored rubric item on the examination and it is easy to lose. |
| <b>Two distributions are described separately rather than compared.</b>        | Use explicitly comparative words — higher, more variable, more skewed. Separate descriptions score as incomplete.                 |
| <b>The normal model is applied to obviously non-normal data.</b>               | Assess the shape first. The normal model is a choice that must be justified.                                                      |

**TEACHING NOTE**

*This is the highest-weighted unit on the examination and it sets the writing standard for the year. Establish now that every numerical answer is accompanied by a contextual sentence; the standard cannot be retrofitted in April.*

**UNIT 2 · 9-10 LESSONS · APPROX. 5-7% OF THE EXAMINATION****Exploring Two-Variable Data**

Երկչափ տվյալների ուսումնասիրություն

**TOPICS COVERED**

- Introducing statistics: are variables related?
- Representing two categorical variables; two-way tables and segmented bar charts
- Statistics for two categorical variables
- Representing the relationship between two quantitative variables with scatter plots
- Correlation and its properties
- Least-squares regression models
- Analyzing departures from linearity: residuals and residual plots
- Outliers, high-leverage points, and influential observations
- Transformations to achieve linearity

**LEARNING OBJECTIVES**

- 1 Describe a scatter plot by form, direction, strength, and unusual features, in context.
- 2 Interpret slope, intercept,  $r$ , and  $r^2$  in context, using precise wording.
- 3 Use a residual plot to assess whether a linear model is appropriate.
- 4 Distinguish outliers from high-leverage points from influential observations.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Interprets all regression output components in context in 9 of 10 trials.
- Assesses linearity from residual plots correctly in 4 of 5 tasks.
- Distinguishes the three types of unusual point in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                 | RECOMMENDED RESPONSE                                                                                                                |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b><math>r^2</math> is described as the percentage of points on the line.</b> | It is the proportion of variability in the response accounted for by the linear model. The rubric wording is specific; rehearse it. |
| <b>Causation is inferred from a strong regression relationship.</b>           | Only randomized experiments support causation. Name a plausible confounder for every observational claim.                           |
| <b>Slope is interpreted without 'predicted' or 'on average'.</b>              | The model predicts an average change. The wording is scored.                                                                        |

**TEACHING NOTE**

*Interpretation wording in this unit is scored to a specific standard. Give the student the exact rubric phrasings for slope, intercept,  $r$ , and  $r^2$ , and require them verbatim until they are automatic.*

**UNIT 3 • 11-12 LESSONS • APPROX. 12-15% OF THE EXAMINATION****Collecting Data**

Տվյալների հավաքագրում

**TOPICS COVERED**

- Introducing statistics: do the data we collected tell the truth?
- Introduction to planning a study
- Random sampling and data collection
- Potential problems with sampling: undercoverage, nonresponse, response bias
- Introduction to experimental design
- Selecting an experimental design: completely randomized, blocked, matched pairs
- Inference and experiments: what conclusions are warranted
- Scope of inference: generalization and causation

**LEARNING OBJECTIVES**

- 1 Identify a sampling method and describe how to implement it correctly.
- 2 Design an experiment specifying control, randomization, and replication.
- 3 Explain when blocking is appropriate and describe a blocked design.
- 4 State the scope of inference for a described study precisely.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Describes sampling and experimental procedures completely enough to be carried out, in 4 of 5 tasks.
- Designs valid experiments in 4 of 5 tasks.
- States scope of inference correctly in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                             | RECOMMENDED RESPONSE                                                                                                                              |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Random assignment and random selection are conflated.</b>              | Selection permits generalization; assignment permits causal claims. Both are required for a study that does both, and they are separately scored. |
| <b>A design is described too vaguely to be implemented.</b>               | The rubric requires a description someone could actually carry out. Name the randomization mechanism explicitly.                                  |
| <b>Blocking is applied to a variable of no relevance to the response.</b> | Block on a variable expected to affect the response, and say why.                                                                                 |

**TEACHING NOTE**

*Free-response questions in this unit ask the student to describe a procedure, and vague descriptions score poorly even when the idea is right. Practice writing implementable instructions, not summaries.*

## UNIT 4 • 14-15 LESSONS • APPROX. 10-20% OF THE EXAMINATION

**Probability, Random Variables, and Probability Distributions**

Հավանականություն, պատահական մեծություններ և բաշխումներ

**TOPICS COVERED**

- Introducing statistics: random and non-random patterns
- Estimating probabilities using simulation
- Introduction to probability
- Mutually exclusive events
- Conditional probability
- Independent events and unions
- Introduction to random variables and probability distributions
- Mean and standard deviation of random variables
- Combining random variables
- Introduction to the binomial distribution
- Parameters for a binomial distribution
- The geometric distribution

**LEARNING OBJECTIVES**

- 1 Design and carry out a simulation to estimate a probability.
- 2 Compute conditional probabilities and test independence formally.
- 3 Compute the mean and standard deviation of a random variable and of a combination.
- 4 Recognize a binomial or geometric setting by verifying its conditions.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Designs valid simulations in 4 of 5 tasks.
- Computes conditional probabilities and tests independence at 92% accuracy.
- Verifies binomial and geometric conditions in writing in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                | RECOMMENDED RESPONSE                                                                                          |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Mutually exclusive and independent are treated as the same.</b>           | Mutually exclusive events with non-zero probability are dependent. Construct both examples in one lesson.     |
| <b>Standard deviations are added when random variables are combined.</b>     | Variances add for independent variables; standard deviations do not. Work in variance and convert at the end. |
| <b>Binomial conditions are not verified before the distribution is used.</b> | Fixed $n$ , two outcomes, constant $p$ , independence. Verify all four in writing — this is scored.           |

**TEACHING NOTE**

*Simulation is a scored skill in its own right and appears on free-response sections. Require a described procedure — what represents what, how many trials, what is recorded — not merely an answer.*

**UNIT 5 • 10-11 LESSONS • APPROX. 7-12% OF THE EXAMINATION****Sampling Distributions**

Ընտրանքային բաշխումներ

**TOPICS COVERED**

- Introducing statistics: why is my sample different from yours?
- The normal distribution revisited
- The Central Limit Theorem
- Biased and unbiased point estimates
- Sampling distributions for sample proportions
- Sampling distributions for differences in sample proportions
- Sampling distributions for sample means
- Sampling distributions for differences in sample means
- Conditions for approximate normality

**LEARNING OBJECTIVES**

- 1 Explain what a sampling distribution is and describe its shape, centre, and spread.
- 2 State the Central Limit Theorem precisely and identify when it applies.
- 3 Verify the conditions for approximate normality of a sampling distribution.
- 4 Distinguish the distribution of the population, the sample, and the sample statistic.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Distinguishes the three distributions correctly in 9 of 10 trials.
- Verifies normality conditions in writing in 9 of 10 trials.
- Computes sampling distribution parameters at 92% accuracy.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                       | RECOMMENDED RESPONSE                                                                                                                       |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The Central Limit Theorem is applied to the population or to the sample data.</b>                | It describes the distribution of the sample statistic. Simulate from a skewed population and display all three distributions side by side. |
| <b>Conditions are omitted because the calculation still produces a number.</b>                      | The conditions determine whether the number means anything, and they are scored separately from the computation.                           |
| <b>The standard deviation of the sampling distribution is confused with that of the population.</b> | The sampling distribution's spread shrinks as $n$ grows. Compute both and compare.                                                         |

**TEACHING NOTE**

*This unit is the conceptual hinge of the course: everything before it is description, everything after is inference. A simulated sampling distribution built with the student is the single most effective teaching device available here.*

**UNIT 6 · 12-13 LESSONS · APPROX. 12-15% OF THE EXAMINATION****Inference for Categorical Data: Proportions**

Եզրակացություն կատեգորիկ տվյալների համար. համամասնություններ

**TOPICS COVERED**

- Introducing statistics: why be normal?
- Constructing a confidence interval for a population proportion
- Justifying a claim based on a confidence interval
- Setting up a test for a population proportion
- Interpreting p-values
- Concluding a test for a population proportion
- Potential errors when performing tests; Type I and Type II errors and power
- Confidence intervals for the difference of two proportions
- Justifying a claim about the difference of two proportions
- Setting up and concluding a test for the difference of two proportions

**LEARNING OBJECTIVES**

- 1 Construct a confidence interval after verifying all conditions and interpret it in context.
- 2 State hypotheses in context and carry out a test for a proportion.
- 3 Interpret a p-value correctly, avoiding the standard misstatements.
- 4 Describe Type I and Type II errors in the context of a specific study, with consequences.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Verifies conditions in writing before every procedure in 9 of 10 tasks.
- Interprets p-values correctly in 9 of 10 trials.
- Writes conclusions with explicit evidence linkage in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                            | RECOMMENDED RESPONSE                                                                                                                                 |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The p-value is interpreted as the probability the null hypothesis is true.</b>        | It is the probability of a result at least this extreme assuming the null is true. Rehearse the correct sentence aloud in every lesson of this unit. |
| <b>Failing to reject is reported as proving the null true.</b>                           | Write 'we do not have convincing evidence that...' rather than 'there is no difference'.                                                             |
| <b>A conclusion is stated without linkage to the p-value and the significance level.</b> | The rubric requires the comparison to be stated. 'Because the p-value of 0.03 is less than $\alpha = 0.05$ , we reject...' is the scored form.       |



**TEACHING NOTE**

*The four-step template — state, plan, do, conclude — becomes mandatory here and applies to every inference procedure for the rest of the course. Introduce it as a permanent structure, not as a Unit 6 device.*

**UNIT 7 • 13-14 LESSONS • APPROX. 10-18% OF THE EXAMINATION****Inference for Quantitative Data: Means**

Եզրակացություն քանակական տվյալների համար. միջիններ

**TOPICS COVERED**

- Introducing statistics: should I worry about error?
- Constructing a confidence interval for a population mean
- The t-distribution and why it is needed
- Justifying a claim about a population mean based on a confidence interval
- Setting up and concluding a test for a population mean
- Confidence intervals for the difference of two means
- Setting up and concluding a test for the difference of two means
- Inference for paired data and matched pairs
- Skills focus: selecting an appropriate inference procedure

**LEARNING OBJECTIVES**

- 1 Select the correct procedure among one-sample, two-sample, and paired designs.
- 2 Construct and interpret confidence intervals for means after verifying conditions.
- 3 Carry out and conclude significance tests for means.
- 4 Recognize paired data and explain why an independent-samples procedure would be wrong.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Selects the correct procedure in 9 of 10 scenarios.
- Verifies conditions and constructs intervals at 92% accuracy.
- Writes complete conclusions with evidence linkage in 9 of 10 trials.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                    | RECOMMENDED RESPONSE                                                                                                      |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Paired data is analyzed with a two-sample procedure.</b>                      | Look for a natural pairing — same subject, matched pair, before and after. Identify the design before selecting anything. |
| <b>The z-procedure is used for means when <math>\sigma</math> is unknown.</b>    | The t-distribution applies whenever the population standard deviation is estimated from the sample.                       |
| <b>Conditions are checked mechanically without reference to the actual data.</b> | Examine the sample for skew and outliers. The condition check must reference what was observed.                           |

**TEACHING NOTE**

*Procedure selection is the skill that most distinguishes strong candidates, because the examination provides the scenario and not the procedure. Devote at least one full lesson to selection alone, with no computation permitted.*

**UNIT 8 · 8-9 LESSONS · APPROX. 2-5% OF THE EXAMINATION****Inference for Categorical Data: Chi-Square**

Եզրակացություն կատեգորիկ տվյալների համար. խի-քառակուսի

**TOPICS COVERED**

- Introducing statistics: are my results unexpected?
- The chi-square test for goodness of fit
- Expected counts in two-way tables
- The chi-square test for homogeneity
- The chi-square test for independence
- Skills focus: selecting an appropriate inference procedure for categorical data

**LEARNING OBJECTIVES**

- 1 Distinguish goodness of fit, homogeneity, and independence by the study design.
- 2 Compute expected counts and verify the large-counts condition.
- 3 Carry out a chi-square test and interpret which cells contributed most to the statistic.
- 4 State a conclusion in context with explicit evidence linkage.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Selects the correct chi-square procedure in 9 of 10 scenarios.
- Computes expected counts and verifies conditions at 92% accuracy.
- Interprets cell contributions correctly in 4 of 5 tasks.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                | RECOMMENDED RESPONSE                                                                                                                |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Homogeneity and independence are chosen by the table shape rather than by the design.</b> | Homogeneity compares several populations sampled separately; independence examines two variables in one sample. The design decides. |
| <b>Chi-square is computed on percentages rather than on counts.</b>                          | The test requires observed counts. Convert back if the data is presented as percentages.                                            |

**TEACHING NOTE**

*Although lightly weighted on the multiple-choice section, chi-square appears regularly in free-response and in the investigative task. Do not treat a small percentage as permission to compress the unit.*

## UNIT 9 • 7-8 LESSONS • APPROX. 2-5% OF THE EXAMINATION

**Inference for Quantitative Data: Slopes**

Եզրակացությունն քանակական տվյալների համար. թերությունն

**TOPICS COVERED**

- Introducing statistics: do those points align?
- Confidence intervals for the slope of a regression model
- Justifying a claim about the slope based on a confidence interval
- Setting up and carrying out a test for the slope of a regression model
- Reading and interpreting computer regression output
- Skills focus: selecting an appropriate inference procedure
- Preparing for the investigative task

**LEARNING OBJECTIVES**

- 1 Read a computer regression output table and identify every relevant quantity.
- 2 Construct and interpret a confidence interval for a regression slope.
- 3 Carry out a test for the slope and conclude in context.
- 4 Approach an unfamiliar investigative task by extending known procedures.

**MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE**

- Reads regression output correctly in 9 of 10 trials.
- Performs slope inference with conditions verified at 92% accuracy.
- Produces a reasoned response to an unfamiliar investigative task in 4 of 5 attempts.

**COMMON MISCONCEPTIONS AND TEACHER RESPONSE**

| MISCONCEPTION                                                                                                     | RECOMMENDED RESPONSE                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>The standard error of the slope is confused with the standard deviation of the residuals in output tables.</b> | Practice reading output from several software packages until the layout is not a source of error.                         |
| <b>The investigative task is treated as requiring an unknown procedure.</b>                                       | It requires extending known procedures to an unfamiliar setting. Practice with released investigative tasks specifically. |

**TEACHING NOTE**

*The investigative task is the final free-response question and is weighted more heavily than the others. It rewards a student willing to reason from principles under uncertainty; practice it as its own skill using released tasks.*

## ASSESSMENT AND MASTERY POLICY

Գնահատում և յուրացման քաղաքականություն

| ASSESSMENT                         | WHEN                                   | WHAT IT MEASURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Placement Assessment</b>        | Before enrollment                      | Arithmetic fluency, graph interpretation, and — most importantly — written explanation of a quantitative claim.                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Unit Check</b>                  | End of each unit · 55 min              | AP-format multiple-choice and free-response, scored with official rubrics.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Four-Step Portfolio</b>         | Continuous from Unit 6                 | Every inference procedure written in the state-plan-do-conclude structure; the primary predictor of free-response performance.                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Investigative Task Practice</b> | Monthly from Unit 5                    | Released investigative tasks attempted under time and scored with the official rubric.                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Full Practice Examination</b>   | Three times: mid-year, February, April | Complete timed examinations, scored and analyzed item by item.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Examination Format</b>          | AP Exam · May                          | 3 hours. Section I: 40 multiple-choice questions, 90 minutes, 50% of the score. Section II: 6 free-response questions, 90 minutes, 50% of the score — Part A with 5 questions and Part B with one investigative task weighted more heavily. In 2026 the multiple-choice section is administered digitally in Bluebook and the free-response section is handwritten on paper. A graphing calculator with statistical capability is permitted throughout, and formula sheets and tables are provided. |

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

## PACING OPTIONS

Ուսումնական տեմպ

| TRACK                            | CADENCE                           | EXPECTED COMPLETION                                                              |
|----------------------------------|-----------------------------------|----------------------------------------------------------------------------------|
| <b>Standard AP Year</b>          | 2 lessons/week, 75 min            | Full course September to April with April dedicated to examination preparation.  |
| <b>Accelerated</b>               | 3 lessons/week                    | Full course in approximately 30 weeks.                                           |
| <b>After Statistics (MTH-ST)</b> | 2 lessons/week                    | Approximately 24 weeks, weighted toward Units 6–9 and examination technique.     |
| <b>Support / Co-Seated</b>       | 1–2 lessons/week alongside school | Paced to the student's school AP course, with emphasis on written communication. |

| TRACK                        | CADENCE                 | EXPECTED COMPLETION                                                                              |
|------------------------------|-------------------------|--------------------------------------------------------------------------------------------------|
| <b>Examination Intensive</b> | 3 lessons/week, 8 weeks | February to April: full practice examinations, rubric training, and investigative task practice. |

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

## HOMEWORK AND INDEPENDENT PRACTICE

Տնային աշխատանք և ինքնուրույն պարապմունք

Sixty to seventy-five minutes per lesson. Every numerical answer must be accompanied by a contextual interpretation; a bare number earns nothing, in homework as on the examination. From Unit 6 onward every inference problem must be written in the four-step structure with conditions verified in writing before any computation. Assignments include AP-format multiple-choice sets and at least one rubric-scored free-response part. Students self-score their free responses against the official rubric before submission, which is what teaches them to see their own writing the way a reader does.

## PROGRESS MILESTONES REPORTED TO PARENTS

Ուսումնական նվաճումներ

These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Describes and compares distributions completely and in context.
- ◆ Uses the normal distribution in both directions.
- ◆ Interprets regression output using precise rubric wording.
- ◆ Assesses linearity from residual plots and identifies unusual points.
- ◆ Describes sampling methods and experimental designs implementably.
- ◆ States scope of inference correctly for any design.
- ◆ Applies probability rules and designs simulations.
- ◆ Works with random variables and binomial and geometric distributions.
- ◆ Distinguishes population, sample, and sampling distributions.
- ◆ Constructs and interprets confidence intervals with conditions verified.
- ◆ Carries out significance tests and interprets p-values correctly.
- ◆ Selects the correct inference procedure from a described scenario.
- ◆ Performs chi-square tests and inference for regression slope.
- ◆ Completes an investigative task by extending known procedures.
- ◆ Scored 4 or 5 on a full timed practice examination.
- ◆ Completed AP Statistics — examination ready.

## MATERIALS AND RESOURCES

Նյութեր և ռեսուրսներ

- Graphing calculator with full statistical capability (TI-84 Plus CE recommended).
- College Board Course and Exam Description, formula sheet, and released free-response questions from at least ten years.
- Bluebook practice access for digital multiple-choice familiarity.
- Simulation software for building sampling distributions in Unit 5.
- Վարժարան AP Statistics problem bank — approximately 2,400 items, plus 300 rubric-scored free-response parts and a dedicated investigative task collection.

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